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Analyses of the Barriers and Drivers of Ghana's COVID-19 Vaccination Program Using the 5A Taxonomy: A Scoping Review

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Abstract

The review explored topical vaccination themes in literature to describe the barriers and drivers of effective COVID-19 vaccination in Ghana. We searched Scopus, Medline, Cinahl, and Google Scholar, covering papers published exclusively in Ghana between December 2019 and January 2023. Using predefined inclusion and exclusion criteria, 27 papers were included in the review. We adopted the 5As taxonomy to organize the papers into five dimensions, namely, Access, Affordability, Awareness, Acceptance, and Activation. Our analysis shows that Acceptance was the most frequently research theme ($n = 23$, 85.1%), followed by Awareness ($n = 19$, 70.4%), Activation ($n = 14$, 52%), Access ($n = 7$, 25.7%), and Affordability ($n = 6$, 22.2%). Further, we found awareness campaign messages to be largely concentrated in Accra, Kumasi and other hotspots. Further studies on vaccine financing, access and inclusion of individuals and families in information dissemination campaigns is highly recommended.

Plain language summary

Assessing the challenges and achievements of Ghana's COVID-19 vaccination exercise and drawing insights for effective public health policies in Ghana.

According to the World Health Organization, vaccination is the safest way to curb the spread of COVID-19. Yet, weak governance and health system infrastructures in low and middle-income countries, coupled with burdening socioeconomic effects of the pandemic, had crippled vaccine programs in LMICs. Accordingly, this national-level scoping review sets to understand the topical COVID-19 vaccination themes explored in literature, to identify the barriers and drivers of vaccination in Ghana. We searched Scopus, Medline, Cinahl, and Google Scholar, covering papers published exclusively in Ghana between December 2019 and January 2023. Using predefined inclusion and exclusion criteria, 27 papers were included in the review. We adopted the 5As taxonomy to organize the papers into 5 dimensions: Access, Affordability, Awareness, Acceptance and Activation. Our analysis shows that Acceptance was the most frequently research theme (n = 23, 85.1%), followed by Awareness (n = 19, 70.4%), Activation (n = 14, 52%), Access (n = 7, 25.7%), and Affordability (n = 6, 22.2%). Aside from the limited publications identified on cost and access, we found that albeit awareness campaign was widespread, the campaign messages were largely concentrated in Accra, Kumasi and other hotspots. Additionally, approaches required to engender peer-to-peer education toward vaccine uptake among Ghanaians was absent. Consequently, further studies on vaccine financing, vaccine access, and awareness creation through community leadership is highly recommended. The high concentration of records on Acceptance, Awareness

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and Activation in this study indicates the need for government to pay attention to these areas to ensure effectiveness and efficiency in future vaccination exercises.

Keywords
COVID-19, vaccination, barriers, drivers, scoping

Introduction

By January 31, 2023, the world recorded 754,367,807 COVID-19 infections, with over 6,825,461 deaths and millions of businesses lost (World Health Organization [WHO], 2023). The high fatality and hospitalization rates prompted the rapid development of vaccines to contain the virus (Dinleyici et al., 2021; WHO, 2021a).

During the last quarter of 2020, various vaccines, such as Pfizer-BioNTech, Moderna, AstraZeneca, Sinovac, COVAX, and Johnson and Johnson, among others, were developed (WHO, 2021b). However, cost limited access to the vaccines, and countries' vaccine procurement aligned with their financial capacity (Green and Loualiche, 2021). As of December 1, 2020, a jab of mRNA-1273 (Moderna) was selling for $37, Coronavac (Sinovac) for $30, BNT162b2 (Pfizer & BioNTech) for $20, Gam-COVID-Vac Sputnik V at $10, JNJ-78436735 (Johnson & Johnson) for $10, and ChAdOx1 nCoV-2019 (AstraZeneca) for $4, among others (Statista, 2020, December 1). Estimates from December 2020 indicate that while direct purchase agreements allowed high-income countries to order close to$4 billion worth COVID-19 vaccines, Low and Middle-Income Countries (LMICs) could buy confirmed doses worth only$2.7 billion (Burgess et al., 2021). For instance, before their development, the United States had already ordered Pfizer vaccines worth $1.95 billion totaling 500 million doses, and an additional 300 million doses of Moderna COVID-19 vaccines (Ofori et al., 2023; U.S. Department of Health and Human Services, Public Health Emergency, 2020).On the contrary, due to their limited financial capacity, LMICs relied heavily on the COVAX ACT Accelerator agreement which covered only 20% of their total vaccination needs (Burgess et al., 2021). Economic recessions during the pandemic and unprepared health systems in LMICs hindered the acquisition of vaccines resulting in low vaccination coverage in these countries (Afrifa-Anane et al., 2022). For instance, while the United Arab Emirates and Portugal achieved over 90% COVID-19 vaccination coverage in 2022, Yemen, Haiti, and the Democratic Republic of Congo recorded below 2% (Lupu and Tiganasu, 2023).

Ghana's COVID-19 vaccination experience mirrored that of other LMICs (Yawson et al., 2017). The country has historically relied on donations to fulfill its vaccination needs. Between 2014 and 2019, for instance, Ghana received a total of USD $18,059,296 from the GAVI Alliance (2019) in support of its immunization programs. Following a declining domestic health expenditure of 1.38% as a share of its Gross Domestic Product (GDP) in 2019, vaccine financing remained a significant drawback to Ghana's COVID-19 immunization efforts (Statista, 2023). The National Budgetary Statement on March 12, 2021, revealed a government revenue deficit of GH¢13.6 billion (USD $2,361,111,111), with unplanned expenditures around GH¢11.7 billion (USD $2,031,250,000) (where US$1.00 = GH¢5.76) (Ministry of Finance [MoF], 2021). Furthermore, the Ministry of Finance's March 2, 2021 budget statement indicated a 4.9% shrink in the country's economy since 2020 when the first COVID-19 infection was recorded in the country (Bukari et al., 2021; MoF, 2021). Consequently, like many LMICs, Ghana pinned her hopes on donations for its COVID-19 vaccination needs, and in 2021 became the first African country to receive COVAX vaccines (The United Nations Children's Fund [UNICEF], 2021; WHO, 2021c). On February 24th, 2021, the President received 600,000 COVAX vaccines, aiming to eventually vaccinate 20 million Ghanaians (GAVI Alliance, 2021; WHO, 2021c). However, facing insufficient supplies from GAVI due to the overwhelming impact of the Indian Delta Plus variant, the government of Ghana shifted attention to bilateral and multilateral donor support from Russia, China, European Union, African Union and the United States, among others, to meet its vaccination needs. By January 30, 2023, Ghana administered 23,226,767 vaccine doses to 40.8% of the country's 31 million people, with around 3,580,565 individuals receiving at least one booster dose (Ghana Health Service, 2023).

Although many applauded the 40.8% vaccine coverage as an improvement over Africa's continental average of 15% (World Health Organization, 2022), it remained considerably low compared to richer countries like Palau (115.29%), Malta (108.40%), and Brunei (108.04%), among others, which recorded vaccination rates above the global average (John Hopkins University, 2023). Critics of Ghana's vaccination coverage attributed the country' performance to many factors. According to Botwe et al. (2022), misconceptions of the adverse effects of vaccines on reproductive health, coupled with the unprepared

emergency response system of Ghana's health system, were responsible for the low vaccine coverage (Lamptey et al., 2021). In addition, Forkuo et al. (2022), attributed Ghana's ineffective vaccination efforts to misinformation (54.8%) about the vaccine's efficacy. Out of 382 respondents investigated, about 47.98% believed that vaccines are dangerous (Forkuo et al., 2022), and another 23.98% thought it could worsen allergic reactions. Similarly, Awuni et al. (2022) observed that mistrust for vaccines had negatively influenced intentions to receive COVID-19 vaccines in Ghana. Aside from misconception and mistrust, Afrifa-Anane et al. (2022) noted managerial lapses such as long queues at vaccination centers, misconception, and vaccines shortages undermining what could have been a smoother COVID-19 vaccination campaign in Ghana.

To strengthen Ghana's public health response system to effectively address potential disease outbreaks in the future, a critical analysis of the barriers and drivers of her COVID-19 vaccination exercise using the 5A Taxonomy is timely.

Thomson's 5A Taxonomy

The 5A model is a framework for diagnosing the determinants of vaccination coverage and implications for health policy in specific contexts (Thomson et al., 2016). The framework examines complex bio-psychosocial, organizational, and structural determinants of vaccine coverage under five dimensions: Access, Affordability, Awareness, Acceptability, and Activation. Access removes barriers to vaccination, enabling individuals to reach or be reached by vaccines. Affordability connotes the time and money expended by individuals to receive vaccines. Awareness measures people's knowledge of vaccine benefits and risks. Acceptance assesses the willingness to receive vaccines. Finally, Activation explains the structural preparedness of health systems, and the regularity of reminders to individuals, families, and communities to vaccinate.

The 5A taxonomy is widely adopted in literature. In Crawshaw et al. (2022), the model was used to describe vaccine uptake and hesitancy among migrant populations in Europe. Reviewing 67 records from 16 countries, the paper highlighted access and acceptance as key determinants of COVID-19 vaccine coverage. Similarly, Newman et al. (2022) used the 5A and 5C models to evaluate COVID-19 vaccine hesitancy among marginalized populations in the USA and Canada and observed that 5A has advantages in explaining structural hinderances to vaccine access, including location, time, cost, trust, individual attitude, health beliefs and contact with health systems, among others (Newman et al., 2022).

In Africa, however, the adoption of 5A taxonomy in literature is scarce. A preliminary literature search on Google Scholar database over the past 10 years generated a total of three (3) papers (Kalu et al., 2022; Nyasa et al., 2023; Thomson et al., 2016). Kalu et al. (2022) used 5A taxonomy to conduct a mapping review on COVID-19 vaccine uptake among adults in 33 countries in African countries. The finding of the review of 68 papers found Acceptance to be a principal determinant of vaccination coverage, with minimal influence from Activation. They recommended further studies on the impact of activation. A few limitations were noted in the paper. Most notable was that the review was completed using articles published between 2020 and 2021, and at a time when COVID-19 vaccines were newly developed and deployed. In other words, since the greater part of COVID-19 vaccination occurred between late 2021 and 2023, this constraint affected the number of empirical studies the review could generate on COVID-19 vaccination, and the comprehensiveness of the investigations. In Nyasa et al. (2023), the authors adopted the model to explain health workers' reaction to hepatitis B vaccines in Zambia and found that although the vaccines were Affordable and Accessible to most of the research participants, Awareness of the vaccines was low. They saw the absence r late dissemination of regular notifications (Awareness) for vaccine update to be a major hinderance to Activation. Finally, in a narrative review on the determinants of vaccine uptake using the 5A taxonomy, Thomson et al. (2016) found all the five (5) dimensions to have a notable effect on vaccine uptake. However, they highlighted the significance of the location of vaccination centers, knowledge, perceived safety, past behavior, and the regularity of reminders as imperatives for vaccine uptake. Moreover, majority of African literature adopting the 5A taxonomy targeted multiple countries. This makes it difficult to estimate the actual immunization gaps in individual countries.

In view of these, the review of empirical records on COVID-19 vaccination in specific African countries to inform public health policies is vital.

Methods

Levac et al. (2010) six steps framework for scoping review was employed for this review. These include identifying research questions and linking them to the research purpose; generating literature by comparing the study's feasibility against its breadth and comprehensiveness; using iterative team approaches to screen selections and extract data; chart data to show the numerical summary and themes generated; collate, summarize and report the result; and as an optional step, consult to confirm the findings.

Eligibility Criteria

We searched for empirical records on COVID-19 vaccination in Ghana. Only studies conducted originally in the

Period (2021–2023)	Population	Intervention	Comparator	Outcome
Concepts	Ghana	Vaccination	Treatment as usual	5A Taxonomy
Keywords and phrases	Service provider Service user Ghana health system Policy maker	All brands of COVID-19 vaccines All COVID-19 vaccination centers in Ghana	Treatment as usual	Access Affordability Awareness Acceptability Activation

Source. Authors' conception.

English language and published in peer-reviewed journals were selected. Again, quantitative, qualitative, and mixed-method studies were considered. However, to meet the inclusion criteria, they must be empirical papers conducted between December 1st 2019 and 3rd January 2023.

Search Strategy

We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension for Scoping Reviews (Tricco et al., 2018). As outlined in the standard Joanna Briggs Institute (JBI) systematic review method, we used the following step to search for papers. First, we performed a limited search in the Medline database. The articles we generated were analyzed in Microsoft Excel® (version 2016) to identify keywords and index terms in the titles and abstracts. We then had a discussion on what to include in the final search terms. The outcome of this discussion was further discussed with a public health expert to validate the decision and advise us. Forward, we conducted a full search on the selected databases and used Mendeley Reference Manager (Version 1.19.8) and Microsoft Excel® (2016) sheets to manage the articles generated. Consistent with Webster et al. (2017), we did not consider grey literature for this critical scoping review.

After the review of preliminary literature and consultation with the public health expert, Scopus, Medline, Google Scholar, and Cinahl, databases were searched for relevant articles.

PICO Framework

The study adopted Thomson et al. (2016) definition of the dimensions of 5A taxonomy. Accordingly, Access indicates the removal of barriers to vaccination to ensure that every individual could reach or be reached by COVID-19 vaccines. Affordability denotes the time and money required by households and individuals to take up vaccines. We defined Awareness as the level of knowledge individuals have on the benefits and risks of vaccination. Activation is the readiness of health systems, and strategies adopted to engage individuals, families, and communities to vaccinate. Finally, Acceptance examines the willingness to receive vaccines.

Search String

- ***Scopus***: (TITLE-ABS-KEY (ghana AND covid-19 AND strategy) OR TITLE-ABS-KEY (covid-19 AND vaccination AND ghana) AND TITLE-ABS-KEY (ghana))
- ***Medline***: (ghana covid-19 policy and strategy) AND covid-19 vaccination AND (ghana or ghanaian)
- ***Cinahl***: (ghana covid-19 policy and strategy) AND covid-19 vaccination AND (ghana or ghanaian)
- ***Google Scholar***: (ghana covid-19 policy and strategy) AND covid-19 vaccin* AND (ghana or ghanaian)

Study Selection and Data Extraction

We designed a standardized extraction form based on PRISMA guidelines for the review. Search and screening was carried out by LY and AHNK with guidance from GAA. The search results were imported into Mendeley Reference Manager (version 1.19.8) and screened at three levels. First, we screened the titles, language, and dates of publication. To be included in the review, the title must contain the terms "SARS-Cov-2 or COVID-19 or vaccination or vaccine." Further, only articles originally published in English language between December 2019 and January 2023 were selected. Next, we screened the abstracts to establish their relevance by types of research papers, geolocation and the research methodology used. We selected only empirical studies conducted in Ghana. Finally, a full text review was conducted to determine completeness, and eligibility to be included in the final list. In addition to extracting records based on relevant themes, the extracted materials were also scrutinized against the quality of information presented. At any point where the two authors could not agree on the decisions to exclude or include a paper, the views of GAA, a Health Service Administrator and Public Health expert, were sought.

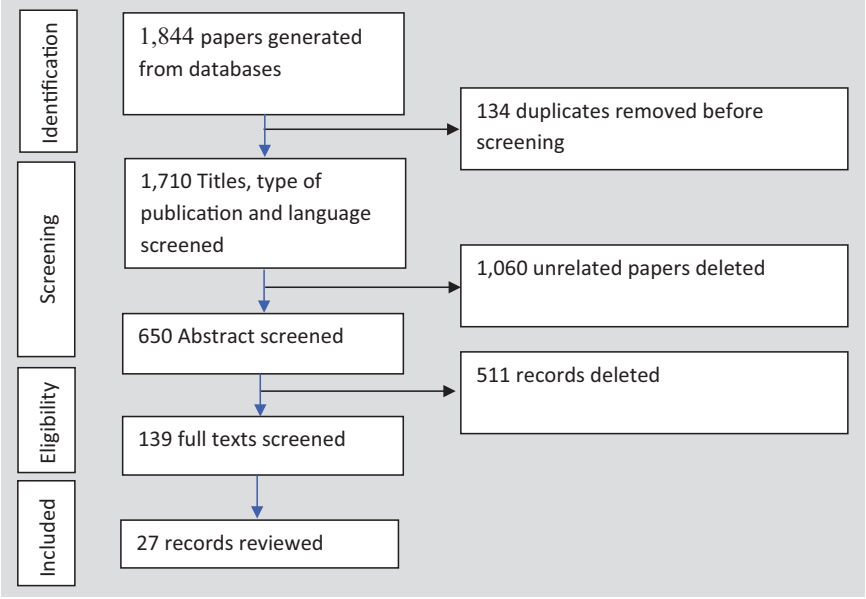

Figure 1. Summary of literature search.

Data Summary

Our combined search from the selected databases generated 1,844 papers. Using authors' names and titles of papers, 134 duplicates were detected and deleted. Together with the title screening, language screening and screening for completeness, an additional 1,060 irrelevant records were deleted. Abstract screening was conducted to determine: (i) the appropriateness of the research population, (ii) method used, and (iii) ensure that the studies were conducted empirically. This led to the deletion of an extra 511 papers. Finally, a full-text screening of 139 records was conducted. From this, we found 27 papers to be eligible for inclusion in the final analyses (Figure 1). The University of Ghana Library System granted access to the databases.

Quality Appraisal and Ethical Considerations

Given that a scoping review is primarily focused on comprehensiveness rather than the quality of the individual papers, the included records were only assessed for their relevance (Brien et al., 2010; Pham et al., 2014).

To avoid publication bias, a well-defined eligibility criteria was used to identify the records, and justifications for the inclusion of records presented and approved before the records were included. In addition, to ensure research integrity, the authors engaged a public health expert to advise them during disagreements.

Collating Results

We adopted Thomson et al. (2016) 5A taxonomy to synthesize the articles under five themes. These include Access, Affordability, Awareness, Acceptance, and Activation (Thomson et al., 2016, p. 1019) (Table 3).

Analysis

LY and HNKA undertook analysis of records. Mendeley reference manager and Microsoft Excel (2016) were used to prepare a literature summary table under relevant themes, namely, authors, year of publication, scope of the study, sector, study objectives, method, research design, sample size (Table 1), title, results, drivers of vaccination and barriers of vaccination (Table 2). The papers were compared against their contents and thematized in line with the 5A model and the research objective. We used narrative review for this paper.

Results

Each of the 27 research papers included in this review was carried out exclusively in Ghana (Table 1). On the research methodology adopted, we found that majority of the records 23 (85.2%) used quantitative research methodology, 2 (7.4%) employed qualitative methodology, and 2 (7.4%) used mixed-research methodology (Figure 2). On the research designs used, majority

Table 1. Authors, year of publication, scope, sector, type of study, method, and design.

Author	Year	Scope	Sector	Method	Design	Sample size	Journal
1) Acheampong et al.	2021	National	PH	Quantitative	Cross-sectional	2,345	Vaccine
2) Addo et al.	2021	National	PH	Quantitative	Cross-sectional	1,768	Human Vaccines & Immunotherapeutics
3) Adjaottor et al.	2022	National	PH	Quantitative	Cross-sectional	817	International Journal of Environmental Research and Public Health
4) Adomako et al.	2021	National	PH	Quantitative	Cross-sectional	613	Ghana College of Pharmacists Journal
5) Afrifa-Anane et al.	2022	Regional	PH	Qualitative	Cross-sectional	30	PLoS ONE
6) Agyekum et al.	2021	National	HHR	Quantitative	Cross-sectional	234	Advances in Public Health
7) Alhassan et al.	2021	National	PH	Quantitative	Cross-sectional	1,556	Tropical Medicine and Health
8) Alhassan et al.	2022	National	HF	Quantitative	Cross-sectional	697	BMC Public Health
9) Amo-Adjei et al.	2022	National	HSP	Mixed Method	Sequential mixed-method investigation	415	Archives of Public Health
10) Amponsa-Achiano et al.	2022	National	HSP	Quantitative	Descriptive	142,4008	Emerging Infectious Diseases
11) Aram et al.	2021	National	PH	Quantitative	Cross-sectional	62	Research Square
12) Asumah et al.	2022	Regional	CC	Quantitative	Cross-sectional	215	Ghana Medical Journal
13) Awuni et al.	2022	National	PH	Quantitative	Cross-sectional	223	PLoS ONE
14) Boi-Dsane et al.	2023	National	CC	Quantitative	Cross-sectional	240	Health Science Reports
15) Botwe et al.	2022	National	Radio.	Mixed method	Cross-sectional	108	Radiography
16) Dubik	2022	Regional	PH	Quantitative	Cross-sectional	420	Risk Management and Healthcare Policy
17) Forkuo et al.	2022	Facility based	PH	Quantitative	An analytical cross-sectional study	382	Ghana Medical Journal
18) Kyei-Arthur et al.	2022	National	PH	Quantitative	Cross-sectional	415	PLoS ONE
19) Lamptey et al.	2021	National	HSP	Quantitative	Cross-sectional	1,000	Clincal and Experimental Vaccine Research
20) Lohiniva et al.	2022	National	Media	Qualitative	Qualitative infodemic management system	31,732,126	JMIR Infodemiology
21) Nonvignon et al.	2022	National	HF	Quantitative	COVID-19 Vaccine Introduction and deployment Costing (CVIC) tool	17,459,408	Vaccine
22) Ofori et al.	2023	National	HSP	Quantitative	Age stratified model	1,000,000	Emerging Infectious Diseases
23) Okai & Abekah-Nkrumah	2022	National	PH	Quantitative	Cross-sectional	362	PLoS ONE
24) Serwaa et al.	2021	National	HHR	Quantitative	Cross-sectional	654	Human Vaccines & Immunotherapeutics
25) Twum et al.	2021	National	Soc. Mkt.	Quantitative	Cross-sectional	478	Journal of Social Marketing
26) Yeboah et al.	2021	Regional	PH	Quantitative	Cross-sectional	1,560	International Journal of Environmental Research and Public Health
27) Zhou et al.	2022	National	Com.	Quantitative	Cross-sectional	3,079	PLoS ONE

Table 2. Factors associated with Ghana's COVID-19 vaccination exercise.

Author	Title	Findings	Driver	Barriers
1) Acheampong et al.	Examining vaccine hesitancy in Sub-Saharan Africa: A survey of the knowledge and attitudes among adults to receive COVID-19 vaccines in Ghana	51% vaccine acceptance among urban adults above 15 year. 21% were unlikely to take the vaccine, while another 28% were undecided. Males (56%) are more likely than females (46%) to accept it. Adults above 55 years are more likely to take the vaccine (66%). The least likely are the 15–25 age group (40%).	Need to protect family, friends, and other people in the community (69%); Prevent COVID-19 infection (67%); As a public health responsibility to help fight the pandemic.	Not sure of clinical safety of the vaccines (61%); Not well informed about the possible effects (53%); Not sure if the vaccine is effective (35%).
2) Addo et al.	Guarding against COVID-19 vaccine hesitance in Ghana: analytic view of personal health engagement and vaccine related attitude Prince	Direct positive relationship between personal health engagement and vaccine attitude and vaccination willingness. COVID-19 susceptivity fear have a significantly positive association with vaccine attitude and vaccination willingness.	Personal health engagement reduces hesitancy. The fear of COVID-19, COVID-19 susceptivity, and personal health engagement produced 56% of the variance in vaccination attitude.	Lack of education on the spread and fatalities of COVID-19.
3) Adjaottor et al.	Predictors of COVID-19 stress and COVID-19 vaccination acceptance among adolescents in Ghana	Females have significantly higher scores on vaccine acceptance. A negative relationship between COVID-19 vaccination acceptance and fear of COVID-19. A significant negative relationship between COVID-19 vaccination acceptance and traumatic stress.	Trust in COVID-19 information. Danger of contamination. Fear and traumatic stress. (Accounted for 6.1% of all possible predictive factors).	Fear, misinformation, lack of trust in authorities.
4) Adomako et al.	COVID-19 vaccine acceptance among the Ghanaian population	At least 74.7% ($n = 458$) were willing to take vaccines if available.	Media campaigns by government agencies such as the National Commission on Civic Education (NCCE) on popular media outlets corrected the misconceptions people had.	Fear of side effects (61, 39.35%). Apathy (34, 21.94%). Claim that vaccine were not tried and tested (28, 18.06%). Quality of vaccines brought to Africa (22, 14.2%).
5) Afrifa-Anane et al.	Facilitators and barriers to COVID-19 vaccine uptake among women in two regions of Ghana: A qualitative study.	Out of 30 participants, 17 vaccinated. Most participants who did not vaccinate were professionals ($n = 7$). The participants who vaccinated were service and sales workers ($n = 7$).	The desire to protect oneself and family against COVID-19. Education on vaccines. Seeing others receive vaccines. Cost-free vaccine.	Long queues at vaccination centers. Fear of side effects. Misconceptions about vaccines. Shortage of vaccines.
6) Agyekum et al.	Acceptability of COVID-19 vaccination among health care workers in Ghana	39.3% of healthcare workers intended to receive the COVID-19 vaccines. 58.1% trusted media information. Majority of healthcare workers (39.8%) do not trust the appropriateness of economic measures taken with COVID-19. 60.7% indicated non-acceptance. Female healthcare workers less likely to accept vaccines. Medical doctors more likely to accept than nurses and midwives.	52.2% of those who had contact with a COVID-19 patient indicated acceptance. 42.3% trusted the authorities. 58.1% trusted information provided by the media. 48.7% trusted healthcare system and hospitals. 45.7% trusted COVID-19 measures implemented.	66.2% of those with no contact with COVID-19 patients ($n = 94$, 66.2%) indicated no acceptance. Concerns about safety of vaccines ($n = 93$, 65.5%). Adverse side effects ($n = 23$, 14.8%). Those with no relatives diagnosed with COVID-19 were not likely to vaccinate.

(continued)

Table 2. (continued)

Author	Title	Findings	Driver	Barriers
7) Alhassan et al.	COVID-19 vaccine hesitancy among the adult population in Ghana: evidence from a pre-vaccination rollout survey	Approximately 60% said they would not participate in a COVID-19 vaccine trial. 65% will take vaccines 69% will recommend it to others. 58% not willing to pay for COVID-19 vaccine.	Residents of urbanized regions Persons with formal education	Fear Safety concerns. Mistrust in state institutions. Uncertainty Political connotations. Spiritual and religious beliefs.
8) Alhassan et al.	Financing COVID-19 vaccination in sub-Saharan Africa: lessons from a nation-wide willingness to pay (WTP) survey in Ghana	697 (44.8%) will receive the COVID-19 vaccine. 55% of respondents expressed willingness to pay for the vaccine. 386 (55%) were willing to pay an average amount of GH¢ 38.00.	More males and persons who have tertiary education indicated willingness to pay for vaccination. Persons willing to participate in COVID-19 vaccine trial or accept the vaccine and recommend it to others were more willing to pay for it.	Females, the uneducated, and those living outside Greater Accra region were unwilling to pay for vaccines.
9) Amo-Adjei et al.	Trust and willingness toward COVID-19 vaccine uptake: a mixed-method study in Ghana, 2021	69% and 98% of respondents who moderately and very much trusted the vaccine were willing to vaccinate. Those who do not trust the vaccine; 73% will not accept it. Health workers (45%) were more inclined to trust vaccines (45%).	Trust in the effectiveness and safety of the vaccine. The professional influence of people in one's social networks. Past triumphs of vaccination programs.	Mistrust in political actors—both local and foreign. Belief in superior protection of God. Misunderstanding of vaccine development processes. A comparatively higher proportion of those working in the informal sector (28%) did not trust vaccines.
10) Amponsa-Achiano et al.	Leveraging lessons learned from yellow fever and polio immunization campaigns during COVID-19 pandemic, Ghana, 2021 Kwame	Capitalizing on existing micro-planning strategies helped reduce the time for vaccine deployment for COVID-19 by 50% and ensured equity in vaccine distribution to rural, urban, and hard-to-reach communities. The cold chain in Ghana was revamped with the support of GAVI. The strategy of using celebrities, media houses, and social media influencers for social mobilization was effective.	Early planning and resource identification. Deployment of human resources and logistics for vaccination campaigns. Deploying surge officers, expanding vaccine coverage, conducting active case searches and social mobilization, and strengthening partnerships with key stakeholders. Using celebrities, media houses, and social media influencers for social mobilization. Collaboration between FDA and EPI ensures a plan for monitoring adverse events after immunization. The cold chain in Ghana was revamped with support from GAVI	Lack of capacity and resources to conduct clinical trials. Ultra-cold chain capacity not decentralized. Capacity was only at the central and regional level and few selected facilities. Need for additional resources to transport vaccines to other parts of the country while maintaining the cold chain.
11) Aram et al.	Effects of vaccine health and safety perceptions on COVID-19 vaccine Uptake in Ghana: Implications for implementing rollout programs	78.55% indicated willingness for mandatory COVID-19 vaccine uptake. 71.45% indicated willingness for voluntary COVID-19 vaccine uptake. The belief that vaccine is healthy and having confidence in vaccine safety is statistically significant with mandatory shot.	80.32% believed vaccines are healthy. 73.06% had confidence in COVID-19 vaccine safety.	81.19% of the respondents were concerned about the source of vaccines. Religion had statistically significant association with taking mandatory COVID-19 vaccine shot.

(continued)

Table 2. (continued)

Author	Title	Findings	Driver	Barriers
12) Asumah et al.	Determinants of COVID-19 vaccine acceptance and hesitancy among healthcare professionals in the Kintampo North Municipality, Bono East Region, Ghana	The overall vaccine acceptance was 78.6% among HCPs, while 21.4% were hesitant to receive COVID-19 vaccines. Majority (57.7%) of HCPs believed that COVID-19 vaccines were safe.	Those who knew someone who has taken the vaccine are 14.9 times more likely to vaccinate. The belief that vaccines are safe and effective in controlling COVID-19 transmission. Those who have never refused vaccines in the past indicated acceptance.	53.0% needed a medical officer's advice before they take the vaccine. 47.4% believed that vaccine convenience (thus, place of vaccination, distance to the vaccine site, cost of the vaccine, etc.) would influence the decision to acceptance. 63.3% concerned about the origin of the vaccine.
13) Awuni et al.	Covid-19 vaccination intensions among literate Ghanaians: Still the need to dissipate fear and build trust on vaccine efficacy?	52.28% of males and 68.57% of females were either indifferent or not willing to vaccinate. More males (48.68%) than females (37.68%) knew about the effectiveness of COVID-19 vaccines. At least 96.08% females and 100% males indicated they heard about the COVID-19 vaccines. About 30.04% believed the vaccines were safe.	Segregating by gender shows 35.29% of males and 18.57% of females agreed that the vaccines were safe. Relatively high level of knowledge about the safety and efficacy of the COVID-19 vaccines. Trust in vaccine safety.	47.98% of respondents believed it was dangerous to vaccinate. 23.98% thought that taking the vaccines could increase allergic reactions. 54.8% did not know about the effectiveness of the vaccines. Mistrust of COVID-19 vaccine was found to have a negative influence on willingness to take vaccines.
14) Boi-Dsane et al.	Prevalence of common adverse events experienced following COVID-19 vaccination and its associated factors in Ghana: Cross-sectional study design.	Country of vaccination, country of origin of AstraZeneca vaccine and age are associated with adverse events of vaccination. Experience of adverse effects was 16 times higher among those who took vaccines in Ghana.	Experience of adverse events was 94% less among those who took AstraZeneca compared to AstraZeneca (Oxford) and 86% less among 40–49 years compared with those who are less than 30 years old.	88.2% had experienced at least one adverse event. The most common adverse event after the first dose was pain at injection site (65.8%), headache (57.5%), tiredness (55.8%), fever (51.7%), chills (39.6%), and muscle pains (38.3%).
15) Botwe et al.	COVID-19 vaccine hesitancy concerns: Findings from a Ghana clinical radiography workforce survey	64 (59.3%) were willing to have the vaccine, but 44 (40.7%) were not.	Belief in vaccine's ability to reduce spread of infection and mortality (54.7%).	Doubts about vaccines' efficacy and side effects (56.8%). Conspiracy theory about vaccines' effects on the Ghanaian race (9.1%). Fertility concerns (4.5%). Only 12% are highly knowledgeable about the efficacy of vaccines.
16) Dubik	Understanding the facilitators and barriers to COVID-19 vaccine uptake among teachers in the Sagnarigu Municipality of Northern Ghana: A cross-sectional study	Intention to receive COVID-19 before its rollout in Ghana was 49%. Intention to accept after rollout was 63%. 52% have completed the COVID-19 vaccination.	Prevention of COVID-19 (84%). To protect families, friends, and community (75%). Protect students or pupils (52%).	Unavailability of the COVID-19 vaccine (52%). Concern about side effects (45%). Inadequate knowledge about COVID-19 vaccines (20%). Only 27% were completely confident in the effectiveness of COVID-19 vaccines.

(continued)

Table 2. (continued)

Author	Title	Findings	Driver	Barriers
16) Forkuo et al.	COVID-19 vaccine acceptance and its determinants in the Bono Region of Ghana	82.6% (242/293) indicated acceptance of the vaccine.	Protection against COVID-19 (96.7%, *n* = 234). Fear of COVID-19' (1.2%, *n* = 3). It compulsory to get vaccinated (2.1%, *n* = 5)	Reasons for refusal include fear of side effects (70.6%, 36/51). Fear that vaccines were made by the Western world to infect them with COVID-19 (56.9%, 29/51). Not susceptible to getting COVID-19 (35.3%, 18/51). COVID-19 is not real, (25.5%, 13/51). Vaccine may be expensive (2.0%, 1/51). Fear of future effects (3.9%, 2/51) and no reason at all (2.0%, 1/51).
18) Kyei-Arthur et al.	Parents' and guardians' acceptability of COVID-19 vaccination for children in Ghana: An online survey	73.3% of parents/guardians would allow their children to be vaccinated. 61% disagreed with the statement "once the vaccine is available and approved, it would be safe."	78.7% of parents and guardians felt COVID-19 could be cured. 53.5% disagreed that "COVID-19 vaccine is the most likely way to stop the pandemic." 48.2% disagreed with the statement, "The best way to avoid the complications of COVID-19 is vaccination." About 67% of the respondents disagreed with the statement "I believe the vaccine program may be likened to the new world order."	Inadequate data about the safety of the vaccine (34.3%). Concern about adverse effects of vaccines (18.0%). Prior adverse reaction to any vaccine (6.3%) Parents and guardians perceiving their children not to be at any considerable risk of developing complications if infected with COVID-19 (6.3%).
19) Lamptey et al.	A nationwide survey of the potential acceptance and determinants of covid-19 vaccines in Ghana	541 (54.1%) respondents would accept vaccines. Trust in the government is strongly associated with vaccine acceptance.	Higher odds of accepting vaccines among participants 18–25 years of age, males, married participants, government workers, and those who did not disclose their income. 907 (90.7%) trust in healthcare system.	Vaccine acceptance by communities threatened by anti-vaccination movements, conspiracy theories, miscommunication, religious dogma, and rumors. Lower odds of accepting the vaccine among participants 36–45 years of age, females, unmarried, bachelors holders, and moderate income earners (2–8 USD or and 8–32 USD per day). 388 (38.8%) had high-risk perception of acquiring COVID-19.

(continued)

Table 2. (continued)

Author	Title	Findings	Driver	Barriers
20) Lohiniva et al.	Infodemic management using digital information and knowledge cocreation to address COVID-19 vaccine hesitancy: Case study from Ghana	A multi-sectoral National Misinformation Task Force was established to implement and oversee the process developed by the Ghana Health Service (GHS).	Establishment of Task force made of Risk Communication and Social Mobilization experts. Involvement public health experts, media, development partners, fact-checkers, UNICEF, and other critical partners.	Through Talkwalker, analysts identified two posts: A local premier league football team coach in which he claimed that the team had lost a game because of their weak physical status after they received the COVID-19 vaccine. And this spread rapidly on social media platforms (Twitter). It came from a national web-based news outlet with a local celebrity, and because football is a popular sport in Ghana and many fans may be influenced by the post. Using Talkwalker, the analysts identified a retweet about a post that claimed to be published by an award-winning doctor who warned that those who take the COVID-19 vaccine will die within 2 years.
21) Nonvignon et al.	Estimating the cost of COVID-19 vaccine deployment and introduction in Ghana using the CVIC tool Justice	The estimated total cost of COVID-19 vaccination ranges between $348.7 and $436.1 million for the target population of 17.5 million. These translate into per person completed primary schedule cost of $20.9–$26.2 and per dose (including vaccine cost) of $10.5–$13.1. The main cost driver is vaccine procurement cost, including shipping (78%) to 83% of the total cost. Human resources for health including vaccinator training, deployment, and compensation, which ranged between 8% and 10% of the total cost. Other minor contributors to cost are cold chain (4%–6%) and data management and monitoring, pharmacovigilance, and oversight (2%–3%).	The least cost drivers were domestic logistics and transport, excluding cold chain, security, and stand-alone technical assistance.	The main cost driver was vaccine doses, including shipping, which accounts for between 78% and 83% of total cost. An estimated 8,437–10,247 vaccinators (non-FTEs) was required during 2021–2022 to vaccinate using a mix of delivery strategies, accounting for 8%–10% of total cost.
22) Ofori et al.	Age-stratified model to assess health outcomes of COVID-19 vaccination strategies, Ghana	Vaccinating the elderly ($\geqslant$65 years of age) could avert >7,000 deaths if 1 million people were vaccinated over 3 months. The number of deaths prevented was the lowest when persons <25 years of age were prioritized in both vaccination time frames (2,317 in 3 months).	Prioritizing the elderly ($\geqslant$65 years of age) resulted in the highest percentage of symptomatic persons (7.19%) at the peak. Vaccinating 1 million persons <25 years of age in 3 months was associated with the lowest percentage (6.75%) of symptomatic persons in the population.	Focusing vaccination initiatives on persons 25–64 years of age was associated with the lowest percentage of symptomatic infections (6.96%)

(continued)

Table 2. (continued)

Author	Title	Findings	Driver	Barriers
23) Okai & Abekah-Nkrumah	The level and determinants of COVID-19 vaccine acceptance in Ghana	62.7% indicated acceptance of vaccines if provided. 37.3% will not vaccinate. Decision to accept the vaccine was influenced by occupation, perceived susceptibility, perceived benefits and attitudes toward vaccines.	Perceived benefit was found to positively influence covid-19 vaccine acceptance. Having a positive attitude toward vaccines increase the likelihood that the individual will accept it.	63.7% were not confident in safety of the vaccines. 23.7% wanted to wait for a while until it seems safe. Perception about vaccines causing sterility (3.7%). Belief that COVID-19 does not exist (3%). Lack of belief in vaccination (5.9%).
24) Serwaa et al.	Non-life-threatening adverse reactions from COVID-19 vaccine; a cross-sectional study with self-reported symptoms among Ghanaian healthcare workers	528 (80.7%) participants experienced adverse reactions, which lasted between 0 and 2 days. The most reported adverse reactions were general body weakness 434 (32.0%), headache 371 (27.3%), and fever 257 (19.1%). Health workers aged 35–39 and 40–44 years had lower odds of adverse reactions compared with those aged 25–29 years.	If Healthcare workers are more educated, have gained more knowledge and understand the relevance and practice of a healthy lifestyle on human health and consequently less likely to suffer from chronic illness. Taking pain medications like paracetamol, before the vaccine administration to avoid unwanted adverse reactions.	High prevalence of adverse reactions among healthcare workers. The most reported systemic adverse reactions were general body weakness, headache, and fever. The probability of female health care workers receiving the COVID-19 vaccines if available was less compared with males.
25) Twum et al.	Intention to vaccinate against COVID-19: a social marketing perspective using the Theory of Planned Behaviour and Health Belief Model	397(83%) indicated they would receive vaccines. 81 (17%) indicated they would not receive vaccines. 54.1% Christians and 28.9% Muslims had shown willingness. Social norm as a predictor of behavioral intention to receive COVID-19 vaccination.	Perceived susceptibility and cues to action encouraged behavioral intention to vaccinate. Attitude toward vaccination formed through affective and cognitive feelings influenced behavioral intention to receive vaccinations.	Perceived barriers have a negative effect on behavioral intention. These include perceived severity, perceived benefits and perceived barriers.
26) Yeboah et al.	To establish a possible link between the knowledge of the disease and compliance with preventive measures, including vaccination	Two thirds (31%) were unwilling to follow basic preventive measures. 35.3% were willing to be vaccinated. Neither knowledge of COVID-19 nor the socio- demographic characteristics had any meaningful influence on preventive measures.	An adequate level of knowledge of COVID-19 (69.9%) was reported. Education and occupation also increased the explained difference in knowledge of COVID-19 from 65.4% to 70.3%.	Adequate knowledge does not translate into a general compliance with the personal preventive public health measures. Misinformation about vaccines and their benefits is common, especially on social media. Widespread recommendation by friends guided the choices of 93.6% of respondents in Ghana on preferences of herbal medicines to used.
27) Zhou et al.	Media attention and vaccine hesitancy: Examining the mediating effects of fear of COVID-19 and the moderating role of trust in leadership	The relationship between media attention and COVID-19 vaccine hesitancy was significantly positive. A significant positive relationship between media attention and fear of COVID-19.	Trust in leadership. Exposure to positive news about COVID-19 in the media.	A positive relationship between fear of COVID-19 and vaccine hesitancy is weakened by people's trust in leadership. Exposure to misleading information on vaccination through media platforms could increase COVID-19 vaccine hesitancy among the rural population.

Table 3. Thomson's 5As Model.

Dimensions	Number observed	Barriers	Drivers	Total
Access	(5), (7), (8), (10), (12), (21), (22)	- Long waiting time - Vaccine shortages - Long distance from vaccination centers - Limited number of vaccination centers due to inadequate human resources and logistics	- Residents near vaccine centers - Using mixed strategies to bring vaccines close to underserved populations - Triaging	7
Affordability	(5), (7), (8), (9), (17), (21)	- High cost of vaccines - High cost of transporting vaccines to centers - The Poor are not willing to pay - Uneducated unwilling to pay - Residents outside Accra unwilling to pay - Unemployed	- Persons with tertiary education qualifications willing to pay for vaccines - Persons who participated in vaccine trials willing to pay for vaccines	6
Awareness	(1), (2), (3), (4), (5), (6), (10), (11), (13), (15), (16), (17), (18), (19), (20), (23), (25), (26), (27)	- Limited or centralized awareness/ education campaign - Fear of side effects - Misconceptions about the vaccines - Fear of vaccine contamination - Misinformation (cultural and social media) - Conspiracy theory (COVID-19 is not real/ COVID-19 can be cured) - Ignorance (public figures)	- Media campaigns - NCCE awareness programs - Trust in vaccination information	19
Acceptance	(1), (2), (3), (4), (5), (6), (7), (9), (11), (12), (13), (14), (15), (16), (17), (18), (19), (22), (23), (24), (25), (26), (27)	- Safety/fear/uncertainty - Side effects - Vaccine effectiveness/perceived benefits - Trust in vaccines - Lack of trust in authorities - Lack of trust in vaccine source - Religious beliefs - Vaccine politics - Working in the informal sector - Females	- To protect oneself and family - Traumatic stress of infection - Seeing others receive the vaccine - Trust in authorities - Persons with no formal education - Persons who had never refused vaccines in the past - Good attitude toward vaccine - Males, aged	23
Activation	(1), (2), (4), (6), (10), (11), (12), (17), (19), (20), (21), (22), (24), (25)	- Mistrust of vaccines - Apathy - No contact with COVID-19 patients - Lack of capacity and resources to conduct clinical trials - Ultra-cold chain capacity centralized - Mandatory vaccination - Absence of medical officers' to advice - Fear of allergic reactions - Well-resourced task force for risk communication and social mobilization - Mistrust in political leadership	- Personal health engagements - Trust in the health system - Trust in COVID-19 protocols - Influence of professionals in one's social network - Early planning and deployment of human resources and logistics - Conducting active case search - Social mobilization and using celebrities and media houses - Collaborations with other state agencies - Good data monitoring - Compensation for frontline workers - Healthcare professionals	14

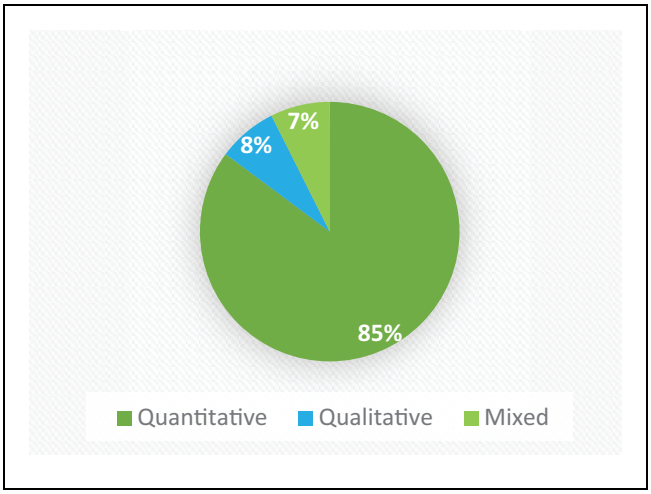

Figure 2. Method.

(n = 22, 81.5%) employed cross-sectional design. The rest include 1 Sequential mixed method design, 1 Descriptive study, 1 QIMS study, 1 Age-stratified model, and 1 CVIC design (Figure 3). In addition, majority of the articles were published in 2022 (n = 15), followed by those published in 2021 (n = 10) and the publications in 2023 (n = 2) (Figure 4). We found most of the articles (n = 14) to be focused on Public Health issues (Figure 5). The remaining 15 were focused on Health communication (n = 3), Radiography (n = 1), Clinical Care (n = 2), Health Financing (n = 2), Health Human Resource (n = 2), and Health Systems Policy (n = 3) (Figure 4). On the scope of the studies, 4% were facility based, 15% were focused on the administrative regions of Ghana, and 81% on the entire country (Figure 6). When mapped against the 5A model, we found only one paper exclusively on Acceptance. Thus each of the remaining 26, (96.3%) articles was focused on multiple dimensions of the model. Specifically, seven articles (25.7%) were related to Access, 6 (22.2%) Affordability, 19 (70.4%) Awareness, 23 (85.1%) Acceptance, and 14 (52%) Activation (Table 3; Figure 7). We presented the summary report for records included in this review in Table 2.

Discussion

The 5A taxonomy is a widely accepted framework for analyzing vaccination coverage (Nyasa et al., 2023; Thomson et al., 2016). In this review, Thomson et al. (2016) 5A taxonomy was adopted to examine empirical literature on COVID-19 vaccination in Ghana. The articles were mapped against five dimensions of the framework to determine gaps in the country's vaccination coverage. The dimensions include Access, Affordability, Awareness, Acceptance, and Activation. From the mapping, we observed that majority of the dimensions overlapped. For instance, articles mapped against Acceptance also assessed Awareness. Similarly, articles assessing Access also measured Affordability.

Acceptance

We found that Acceptance is significantly associated with vaccination coverage in Ghana. Aligned with the narrative review by Thomson et al. (2016), Acceptance emerged the most significant factor influencing vaccination coverage in Ghana. From the records, 23 (85.2%) articles were centered on COVID-19 vaccine Acceptance in Ghana. Aside from their areas of concentration, we analyzed the findings presented in each article and found that the majority reported high vaccine acceptance rates in Ghana. On Table 2, the records show that between 39.3% and 82.6% of study participants were willing to accept COVID-19 vaccines in Ghana. However, the degree of acceptance is influenced by several factors. Some of these include the present health conditions, prior exposure to COVID-19 virus, and societal influencers, among others (Acheampong et al., 2021; Addo et al., 2021; Agyekum et al., 2021; Amo-Adjei et al.,

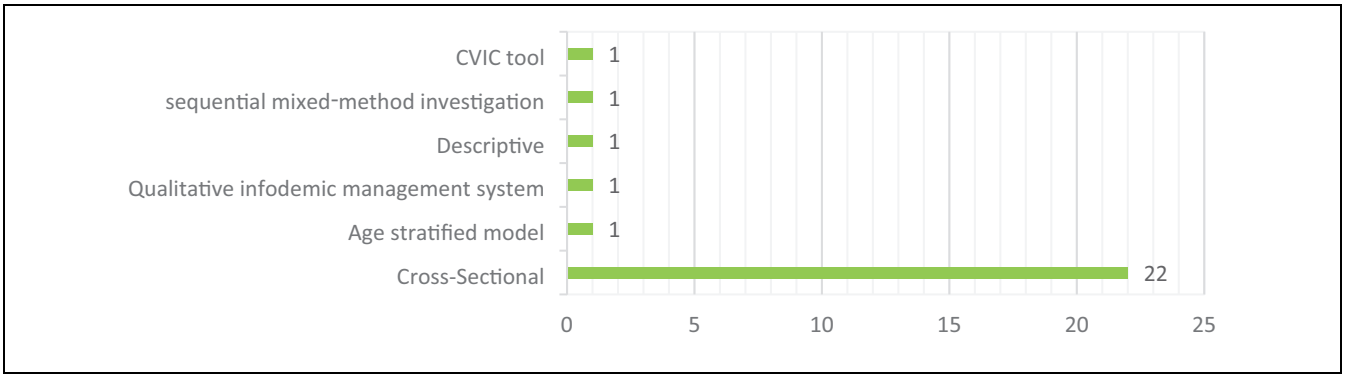

Figure 3. Research design.

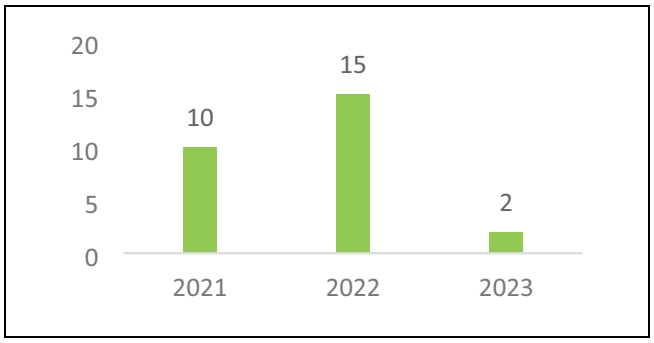

Figure 4. Year of publication.

2022). Acceptance increased among persons with chronic illnesses (Acheampong et al., 2021), individuals infected with COVID-19 (Agyekum et al., 2021), and those influenced by prominent political personalities (Asumah et al., 2022; Lohiniva et al., 2022; Thomson et al., 2016). Additionally, willingness to accept COVID-19 vaccines was high when people are well-informed about their benefits, safety, and associated risks (Adjaottor et al., 2022; Adomako et al., 2021; Agyekum et al., 2021; Kyei-Arthur et al., 2022). As vaccine acceptance is the determinant of herd immunity (Malik et al., 2020) and potentially the most significant indicator for success (Kalu et al., 2022), its not surprising that our review found most of the records to be focused on vaccine acceptance. However, instead of exploring the experiences of research participants, from the research methodologies analyzed, we found majority of these papers to be merely interested in the rates of vaccine acceptance. From the records, 22 out of 23 articles employed quantitative research methodology, and many of them utilizing multiple-item Likert scales (Acheampong et al., 2021; Addo et al., 2021; Adjaottor et al., 2022).

Again, although high acceptance rates were observed among particular populations, including healthcare workers and teachers, the national data shows a low coverage rates in comparison with advanced countries. COVID-19 vaccination uptake in high-income countries generally showcased higher rates of vaccine acceptance, often exceeding the global average in some regions (John Hopkins University, 2023). This is largely because advanced nations implemented robust and extensive vaccination programs, leading to increased awareness and confidence in the vaccines offered (Awuni et al., 2022). Countries such as the USA and UK had invested substantially into education and awareness creation, expanded access to healthcare, and built trust in government policies and healthcare institutions (Deb et al., 2023). In contrast, our review revealed the lack of structural and policy preparedness in Ghana to respond to the pandemic (Kyei-Arthur et al., 2022; Ofori et al., 2023).

In summary, while the factors influencing acceptance are well documented in literature, there remains a need for more comprehensive studies elucidating strategies to enhance vaccine acceptance in Ghana. Additionally, the reliance on quantitative methods warrants exploration into qualitative approaches to provide deeper insights into the nuances of vaccine acceptance among Ghana's populace. Further research could adopt qualitative research designs to explore interventions targeting these factors to bolster acceptance rates and inform robust vaccination strategies.

Awareness

Out of 23 articles concerned with Acceptance, 17 were connected with Awareness. Authors, including Zhou

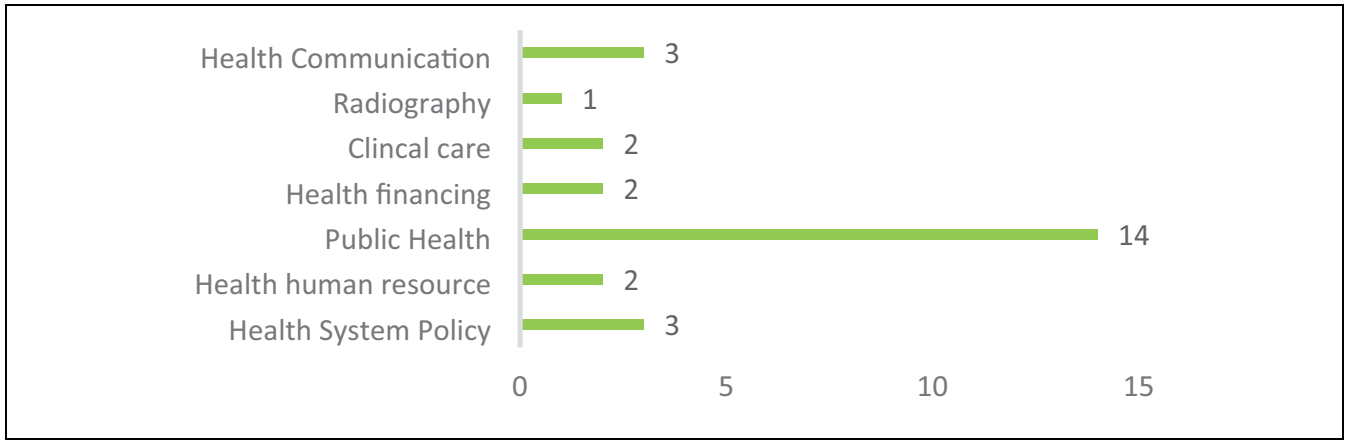

Figure 5. Sector.

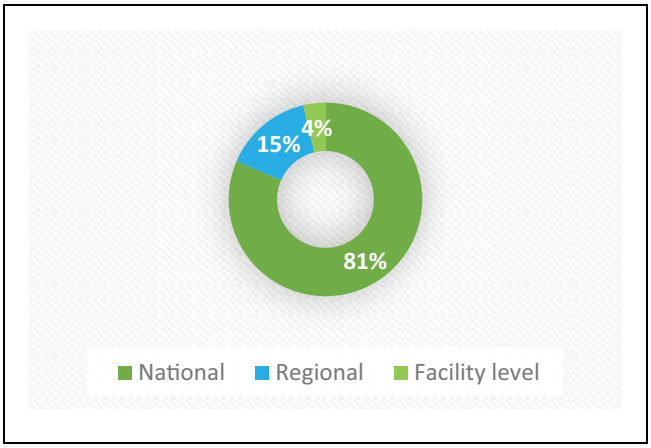

Figure 6. Scope.

et al. (2022), Okai and Abekah-Nkrumah (2022), Boi-Dsane et al. (2023), Kyei-Arthur et al., 2022, and Lohiniva et al. (2022) argued that adequate knowledge of the benefits and risks of vaccination is an indispensable requirement for enhanced vaccination coverage in Ghana. Analyses of these records show high prevalence of misinformation, misconception, and ignorance with associated high vaccine hesitancy in Ghana (Twum et al., 2021; Kyei-Arthur et al., 2022; Yeboah et al., 2021; Zhou et al., 2022). For instance, Okai and Abekah-Nkrumah (2022) noted Ghanaians' concerns about potential risks of vaccine uptake to their reproductive health. Similarly, some Ghanaians expressed fear of vaccines from Western countries, suspecting them of being designed to harm the black race and older persons (Boi-Dsane et al., 2023). Lohiniva et al. (2022) recounted an incident where a local Premier League football coach attributed his team's loss to players' weakened physical condition post COVID-19 vaccination; illustrating the limited understanding many Ghanaians had regarding vaccine benefits and risks. Similarly, Zhou et al. (2022) highlighted that exposure to misleading public communications via media platforms heightened vaccine hesitancy among Ghanaians. It is important to mention, however, that although records from our review supported this finding, contrasting these findings with COVID-19 vaccination statistics from high-income countries reveals a stark divergence in the factors influencing hesitancy. While misinformation and misconceptions play a role globally, the specific concerns regarding the origin and intent of vaccines seem more prevalent in the Ghanaian context compared to high-income countries (Agyekum et al., 2021; Alhassan et al., 2021). Additionally, the impact of influential figures attributing sports outcomes to vaccination effects is not a commonly reported issue in high-income countries.

On the other hand, Yeboah et al. (2021) challenged the notion that greater vaccine knowledge directly translates to increased vaccine uptake. Analyzing 1,560 respondents, they discovered that although over 69.9% were well-informed about vaccine benefits and risks, only 31% were willing to be vaccinated. They argue that extensive knowledge doesn't necessarily result in compliance with preventive measures. Yeboah et al. (2021) admitted, however, that social media and informal relationships are major influencers of vaccination choices, with 93.6% of willing respondents being influenced by recommendations from friends and family. Thus, the paper observed that instead of focusing resources on the types of information communicated, policy makers should rather pay attention to the medium or channel of communication. However, the dominance of social networks in influencing vaccination decisions appears to be more pronounced in Ghana than in high-income countries, where factors like trust in institutions and access to healthcare might play a more significant role. In addition, the recommendation to center attention on communication mediums rather than specific information types warrants further investigation (Yeboah et al., 2021). While this approach might be relevant in Ghana, it might not fully align with the dynamics in high-income

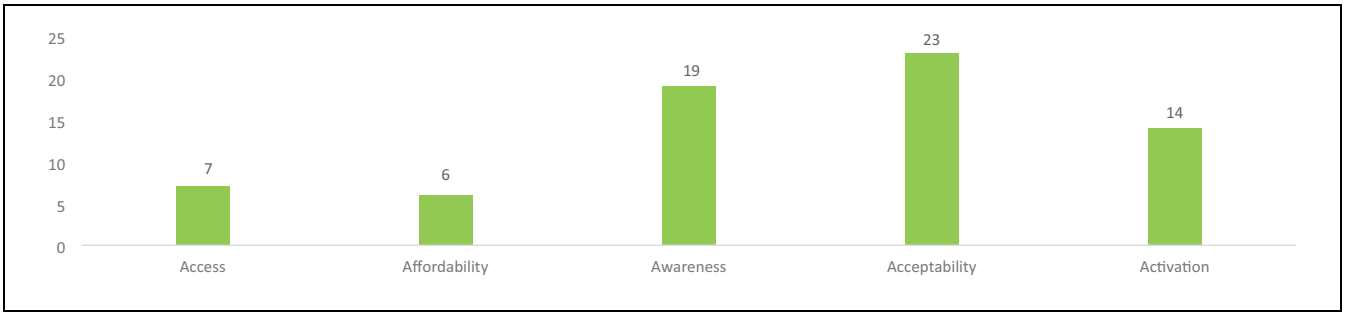

Figure 7. 5A dimensions.

countries where a blend of targeted information and credible sources might play a more critical role in vaccination promotion strategies.

In summary, we are of the view that tailoring communication strategies to suit specific cultural contexts remains crucial, but further research on a blended approach to increase Awareness is necessary to inform vaccination policies in Ghana and other contexts.

Access

Although health systems in LMICs are largely constrained by infrastructure and institutional structures, our review identified only seven papers related to Access dimension with divergent opinions on the phenomenon.

Some of the records reported a positive correlation between selected workplaces and professionals and enhanced vaccine Access (Agyekum et al., 2021; Amo-Adjei et al., 2022; Serwaa, et al., 2021). According to Amo-Adjei et al. (2022) and Agyekum et al. (2021), Ghana's initial vaccine distribution targeted critical patients, the elderly, and frontline workers such as healthcare professionals, security forces, and policymakers (Alhassan et al., 2021, 2022; Aram et al., 2021; Serwaa, et al., 2021). This prioritization minimized travel distances for these groups, provided firsthand vaccine information, and reduced inconveniences during vaccination, potentially fostering higher acceptance. Affordability thus emerged as a significant factor intertwined with access.

In a contrary view, other findings revealed critical structural factors which barred Ghanaians from reaching or being reached by COVID-19 vaccines. Some of these include distance from immunization centers, place of residence, place of work, inequities on vaccine distribution, and vaccine shortages, among others. For instance, in their examination of facilitators and barriers to COVID-19 vaccine uptake among women in two regions of Ghana, Afrifa-Anane et al. (2022) observed that participants could not vaccinate because of long queues at vaccination centers and shortage of vaccines. Queueing and vaccine shortages was particularly frequent when the initial 600,000 jabs were received from the COVAX facility (Afrifa-Anane et al., 2022; Alhassan et al., 2022) and the fear of contracting the disease or dying resulted in a rush for the vaccines. Due to the waiting time required to receive the jabs, many full-time employees were not vaccinated during the initial phase of the rollout.

In comparison with data in high-income countries, we observed stack disparities in resource allocation, distribution strategies, and prioritization. In the US and other high-income countries, equitable distribution was attained due to robust infrastructure, proactive dissemination strategies, and adequate vaccine supply policies (Amponsa-Achiano et al., 2022; Deb et al., 2023; Dubik, 2022). Again, contrary to LMICs where shortages and distance hindered access, high-income countries achieved a broader vaccine accessibility (Dubik, 2022).

The contrast between LMICs and high-income countries underscores the impact of structural inequalities on vaccine accessibility (Ofori, et al., 2023). While high-income nations demonstrated a more systematic approach, LMICs grappled with multifaceted challenges. Addressing these disparities demands a comprehensive approach, considering not only vaccine availability but also equitable distribution strategies, infrastructure improvements, and targeted interventions to mitigate accessibility barriers in LMICs.

Affordability

Affordability describes the monetary and non-monetary expenses related to obtaining COVID-19 vaccines (Thomson et al., 2016). Our findings indicate a close link between affordability and vaccine accessibility with significant effects at the national and household levels (Kalu et al., 2022).

On a national scale, findings from our records show that despite Ghana receiving multiple doses of COVID-19 vaccines from initiatives like COVAX and through bilateral relations with the United States and the European Union, supply was insufficient to cover the entire population. This shortage stemmed from the nation's economic downturn when the pandemic struck, hindering the government's ability to procure the remaining doses. As a result, there was a scarcity of vaccines, limited vaccination centers, and subsequently, reduced access to vaccination (Kalu et al., 2022).

At the individual level, our data reveals a strong association between coverage and vaccine affordability. Household cost elements included service fees, transportation expenses to vaccination sites, and the time invested in waiting for vaccines (Thomson et al., 2016). Alhassan et al. (2021) conducted a study involving 1,556 individuals and found that those residing outside Accra, the capital city, were unwilling to bear the cost of transportation to vaccination centers. This reluctance was likely because populations living outside Accra and other major cities in Ghana fall within the extremely poor category and unlikely to afford the transportation costs associated with receiving the jabs (Alhassan et al., 2021; Amo-Adjei et al., 2022). Furthermore, the long waiting time at vaccination centers acting as a deterrent for workers, especially casual workers and full-time employees (Afrifa-Anane et al., 2022; Thomson et al., 2016). Coupled with limited awareness of the benefits and risks of COVID-19 vaccines outside major cities, it is expected that many

households would lack the motivation spend their time and savings on vaccination.

Comparatively, in advanced countries, accessibility to COVID-19 vaccines tends to be more widespread due to robust healthcare infrastructure and greater financial resources. Affordability barriers are less pronounced, often mitigated through government-funded vaccination programs that prioritize free or low-cost inoculations for the entire population (Amponsa-Achiano et al., 2021; Dubik, 2022; Nonvignon et al., 2022). Moreover, higher awareness levels and better dissemination of information contribute to increased vaccine uptake, minimizing reluctance among individuals and reducing the impact of long waiting times at vaccination sites (Alhassan et al., 2022).

Critically assessing the disparities between high-income countries and regions like Ghana highlights the significant impact of economic constraints on vaccine accessibility. While affluent nations can navigate affordability barriers and ensure comprehensive coverage, lower-income settings face substantial hurdles due to limited resources, ultimately affecting vaccine distribution and uptake. Addressing these disparities demands tailored strategies that account for the economic realities and informational needs of diverse populations.

Activation

Activation is the structural preparedness of health systems, and the effectiveness of reminders for vaccination. We identified fourteen papers showing a push toward both preventive measures and COVID-19 vaccination. For instance, Kalu et al. (2022) and Yeboah et al. (2021) reported that early in the pandemic, emergency strategies and media campaigns, including contact tracing, community address systems for hand hygiene education, hand sanitizers, masking, lockdowns, and social distancing protocols, were implemented to curb community infections and deaths. However, as vaccines became available in greater numbers, emphasis was centered on vaccination awareness campaigns. Media campaigns during the pandemic's early stages played a crucial role in correcting misconceptions and promoting adherence to preventive protocols (Adomako et al., 2021). The media facilitated a strong connection and trust in government-provided information, and compliance with preventive measures. Adomako et al. (2021) documented that these preventive measures were largely successful, especially given the initial ineffectiveness of traditional treatment approaches. Agyekum et al. (2021) reported that 42.3% of respondents trusting government information, for instance, expressed willingness to take the vaccines. Similarly, Addo et al. (2021) found that health personnel engagement increased vaccine acceptance, emphasizing the role of close interactions in reducing hesitancy among health workers. Despite its significance, our records indicate inadequacy in the media awareness campaign (Acheampong et al., 2021). In their study on vaccine hesitancy in Sub-Saharan Africa, Acheampong et al. (2021) revealed that 53% of 2,345 Ghanaian adults surveyed were not well-informed about the vaccines. This lack of information led to 61% expressing doubts about the clinical safety of the vaccines and 35% having doubts about their effectiveness. Addressing this gap in information is vital to overcoming vaccine hesitancy and ensuring widespread acceptance of COVID-19 vaccines in diverse regions.

This contrasts with developed countries where comprehensive communication strategies contribute to higher public awareness and understanding of vaccination efforts. Media campaigns, contact tracing, and community education played a pivotal role in shaping public perception and adherence to safety protocols in countries that recorded high vaccination coverage (Lohiniva et al., 2022; Zhou et al., 2022).

In summary, Activation underscored the importance of early and effective communication strategies, especially in regions with limited information access. However, comparing data on Activation in Ghana with developed countries emphasize the need for tailored approaches to promote public trust. Future efforts should prioritize targeted media campaigns and health personnel engagement to ensure widespread understanding and acceptance of COVID-19 vaccines, contributing to global public health initiatives.

Limitations

The review has some limitations. While our review had successfully assessed operational strategies used to roll out the Ghana's COVID-19 vaccine program, the 5A taxonomy was inadequate in accessing the role of enablers within family units. Some of these enablers include age, gender, rural-urban divides, and other interpersonal relationships which could affect vaccination decisions. While this omission does not take away the relevance of our findings to public health policies in Ghana, we recommend that further research is done in this area to inform the message that media organizations can communicate to the public during a pandemic. Additionally, since our review was focused exclusively on peer-reviewed empirical papers published in English, we are likely to miss out on other information published in editorials, commentaries, and newspaper publications, among others. In the same vein, since many national journals in Africa are not indexed in the databases we used for this research, we could miss useful information contained in papers published in such journals. Finally, we did not consult with stakeholders to validate our

research findings. However, since this process can greatly enhance the quality of the review, we recommend users of our findings to, as much as possible seek additional interpretations from the relevant experts.

Conclusion

There is a plethora of publications on COVID-19 vaccination using different models which are sometimes overlapping. Our work harmonizes all these articles using the 5A taxonomy to scientifically regroup and classify these articles to advise policy makers to enact evidence-informed policies and programs to address the gaps identified. Regarding research gaps, our findings recommend further empirical investigations on vaccine financing (Affordability) and access constrains in Ghana. With many disease outbreaks in Africa, including Ebola, Measles, Hepatitis B and other childhood diseases, vaccination would remain a health policy choice for African nations for many years. The best way to enhance the effectiveness (Acceptance) of vaccination on the continent is to engage in continuous scientific assessment of the strategies (Activation) governments and health entrepreneurs adopt to make the jabs available to those who need it, and the results they produce.

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Data Availability Statement

The data underlying this study, including the literature search records, study selection process, and all data supporting the findings of this review are included in the article.

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